

Yes. I checked the current Intel/Altera F-Tile documentation, and there is strong evidence for the architecture, but I want to distinguish what Intel explicitly states from what we infer from the resource allocation.

1. First: Can we have $2 \times 400\text{G}$ on one F-Tile?

Yes, the resource architecture supports it for the 400GE-8 configuration.

Intel's current F-Tile documentation explicitly lists 400GbE-8 as a supported Ethernet Hard IP mode. It also states that an F-Tile contains up to 16 FGT PMAs + 4 FHT PMAs.

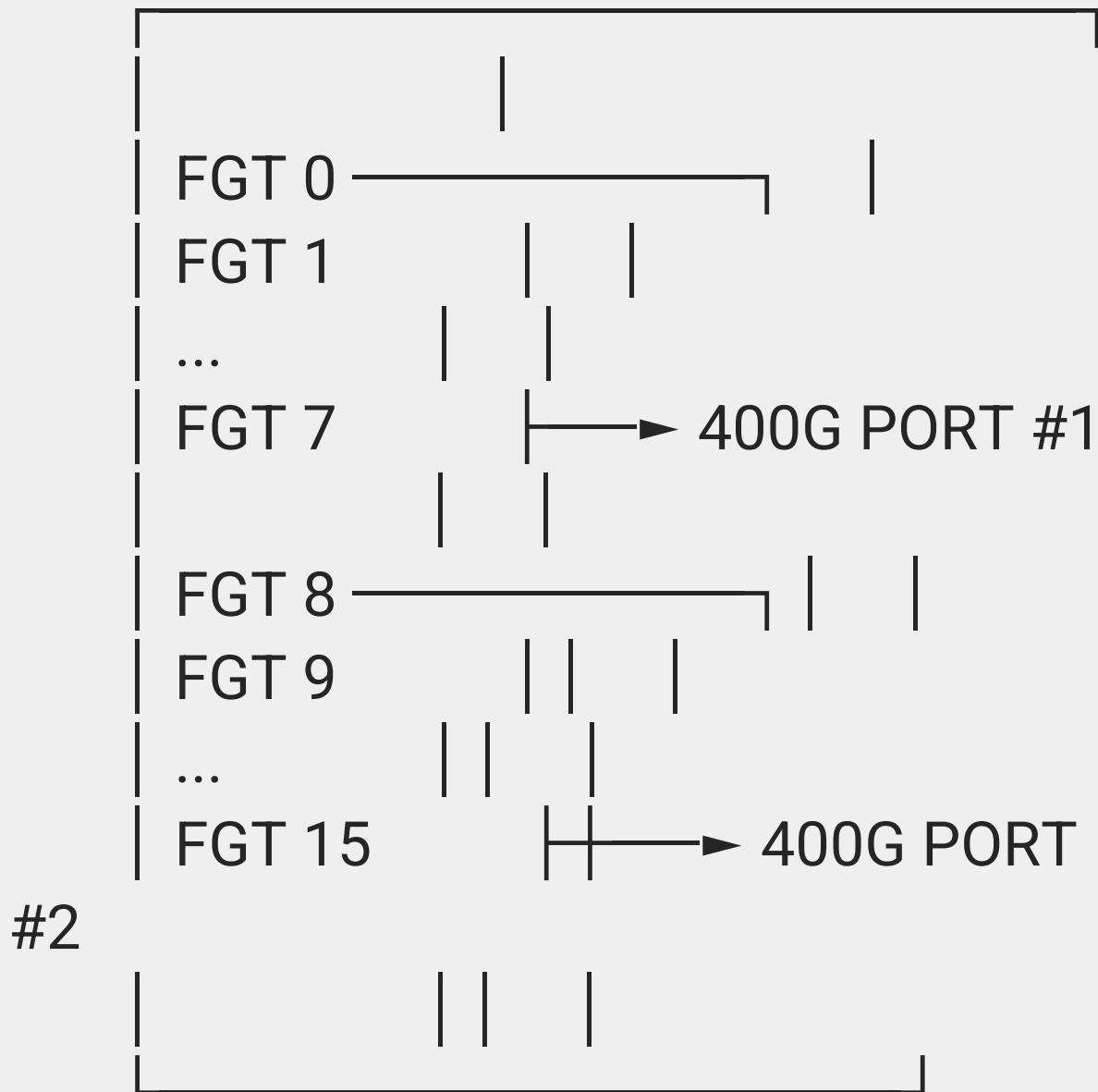
Intel further documents that 400G Hard IP can interface with 16 FGT PMAs.

Therefore:

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ONE F-TILE



Resource-wise:

$1 \times 400\text{GE-8} = 8 \text{ FGT lanes}$

$2 \times 400\text{GE-8} = 16 \text{ FGT lanes}$

F-Tile = up to 16 FGT lanes

So the lane budget fits exactly.

Important qualification

Intel's documentation explicitly confirms 400GE-8 and the 16-FGT resource, but I have not found an Intel sentence/diagram that literally says:

"Two independent 400GE-8 Ethernet Hard IP instances are supported simultaneously on one F-Tile."

So I would classify this as strongly supported by Intel's documented architecture/resource rules, but not the same level of evidence as Intel's explicit 2×100G example. For a production board commitment, I would get the exact OPN/configuration validated by Intel/Altera FAE.

2. Now the System PLL — this is

the important part

The good news is that you don't need one PLL per Ethernet port.

Intel explicitly says:

Each F-Tile system must instantiate one F-Tile Reference and System PLL Clocks IP.

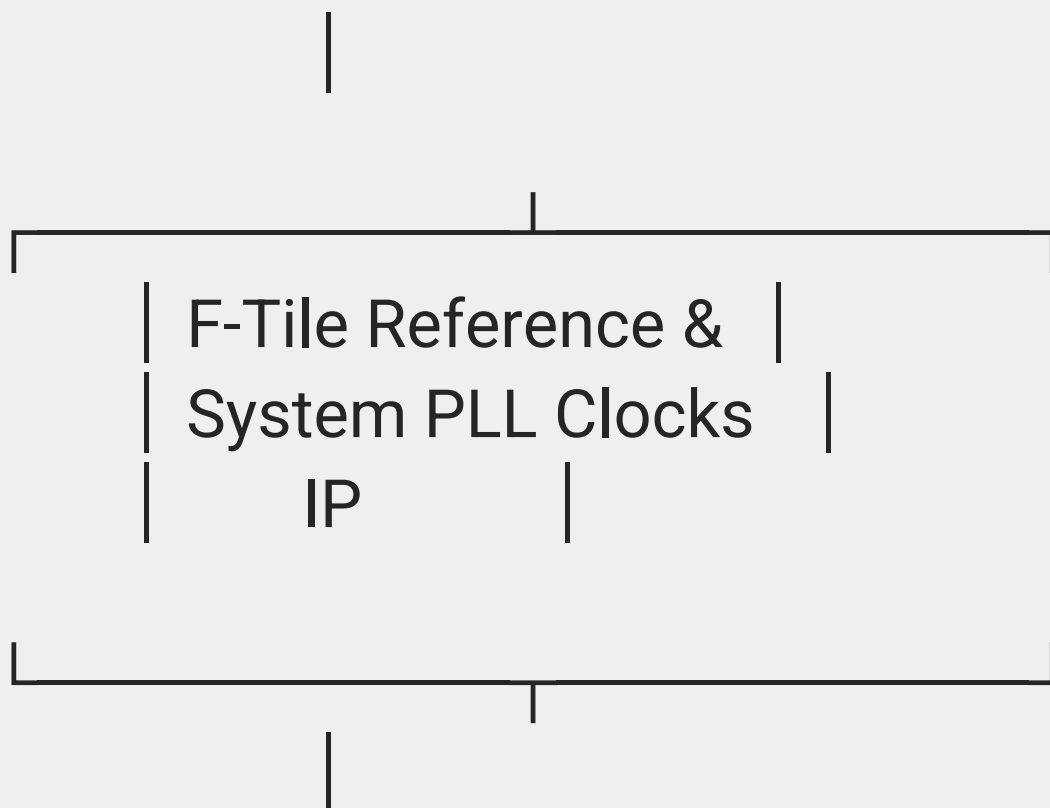
That IP handles the FHT reference clocks, FGT reference clocks, and System PLL configuration.

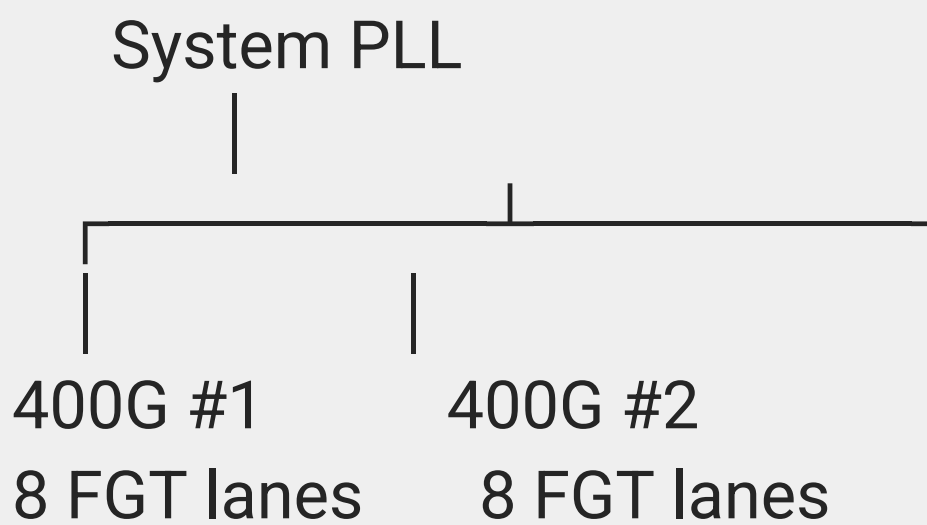
So conceptually:

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F-Tile





The two 400G ports therefore do not require two independent external clock sources.

3. What clock does the 400G Ethernet IP actually use?

This is where the numbers become useful.

For Ethernet modes using RS(544,514) FEC, Intel specifies:

[System PLL \$\geq 830.078125\$ MHz](#)

and the Ethernet IP's corresponding o_clk_pll is:

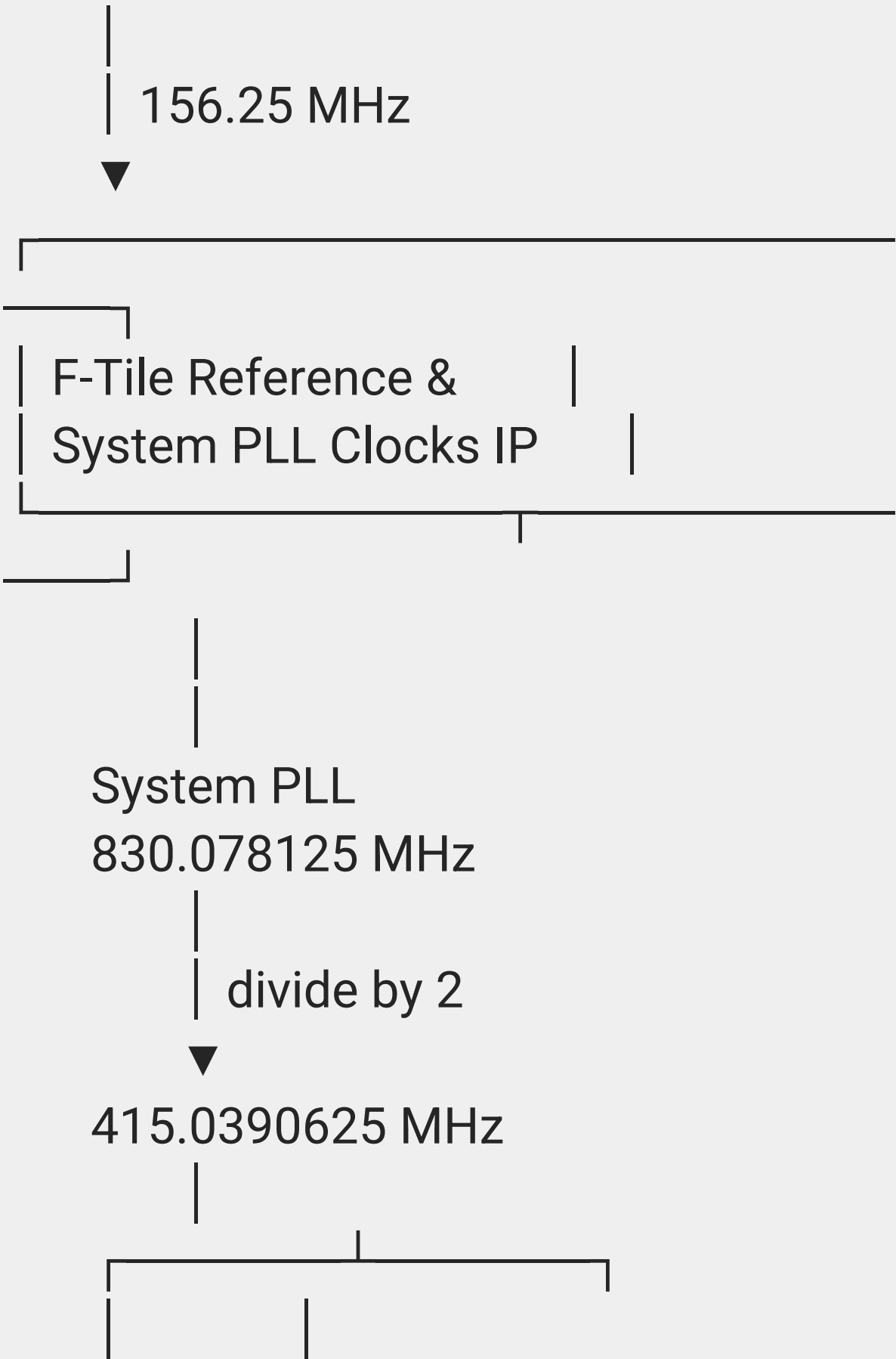
[830.078125](#) / 2 = [415.0390625](#) MHz.

So a typical arrangement is:

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External/reference clock



▼
400G #1

▼
400G #2

Intel's documented preset

ETHERNET_FREQ_830_156 uses:

156.25 MHz reference → [830.078125 MHz](#)

[System PLL output.](#)

4. What about the transceiver reference clock?

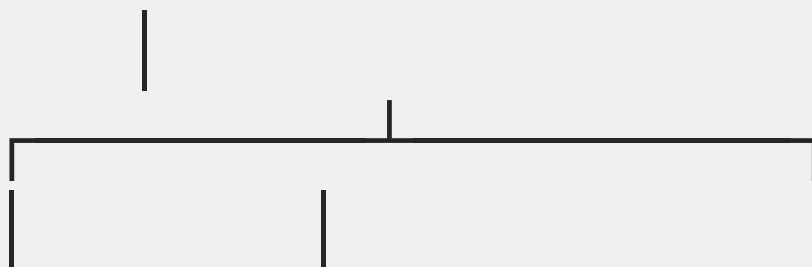
This is different from the System PLL clock.

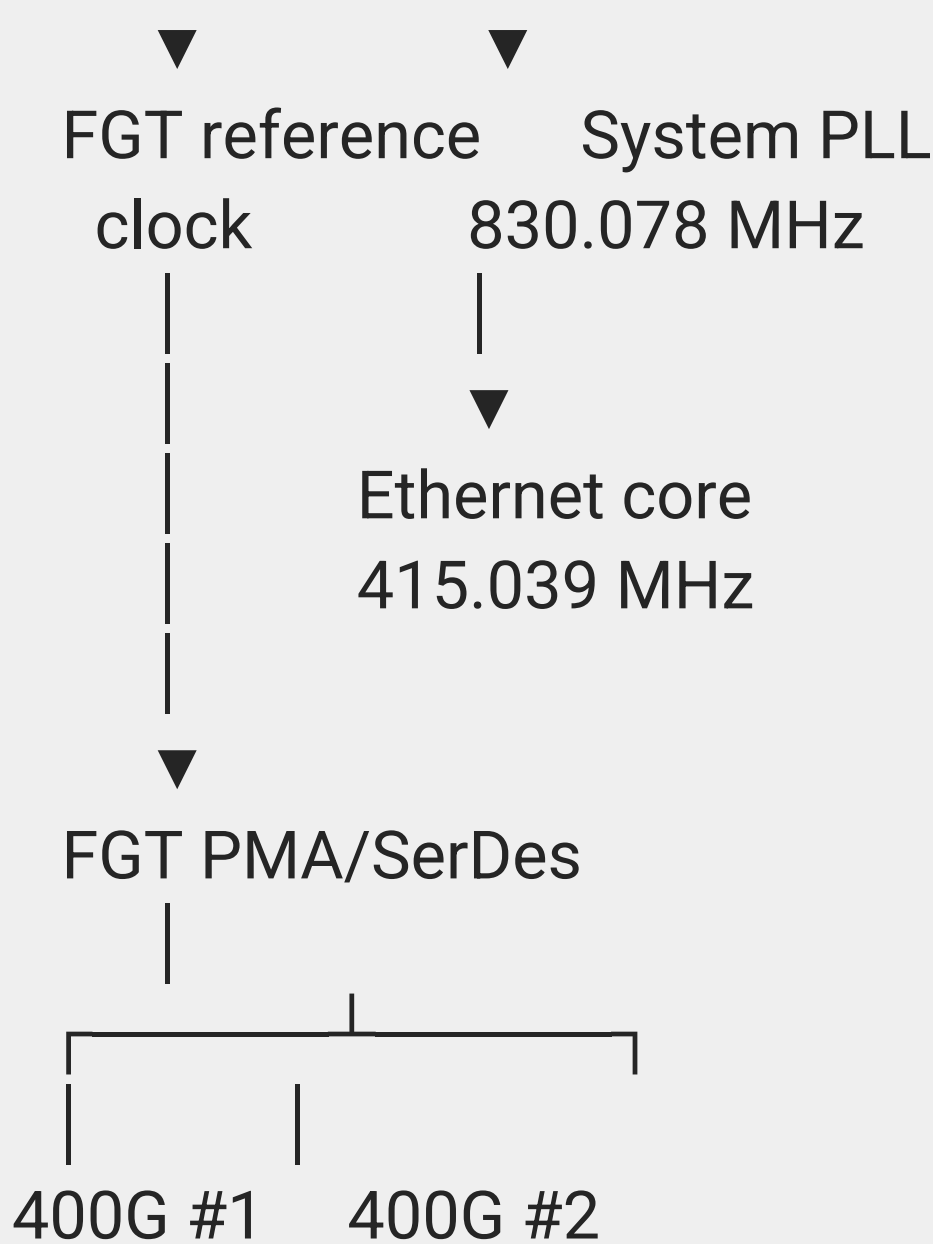
You can think of the clocks as:

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156.25 MHz





Intel specifically documents that the F-Tile Reference/System PLL IP can provide the FGT reference clock and the System PLL clock. And 156.25 MHz is the recommended reference frequency for Ethernet modes; it is the only supported frequency when using FHT or when AN/LT is enabled.

5. What happens to the two 400G ports?

The important thing is that both 400G ports are on the same F-Tile, so their bonded lanes must obey the F-Tile bonding rules.

Intel explicitly says that bonded lanes must:

- be within the same tile

- use the same clock

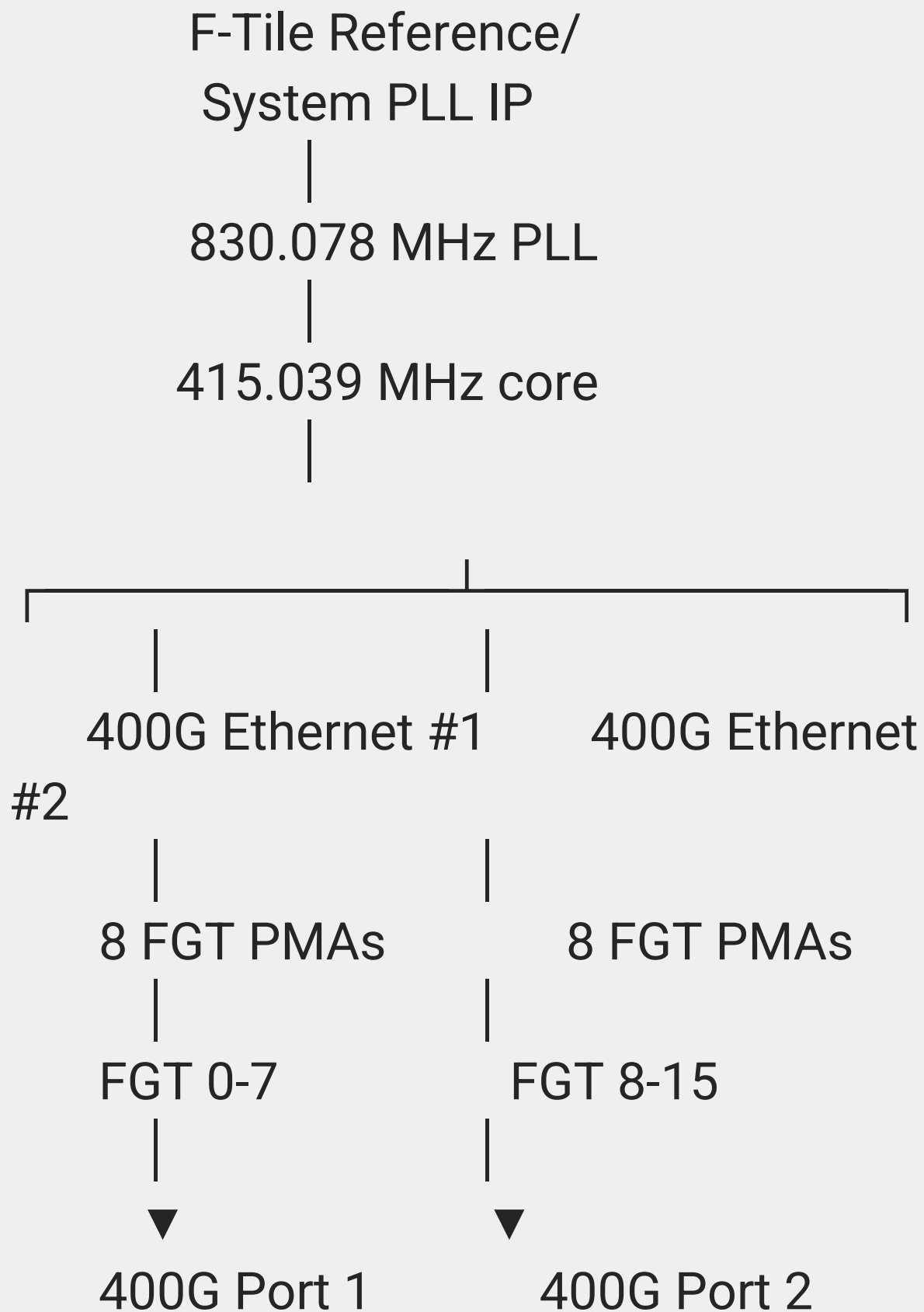
- and for 400G Ethernet Hard IP, the supported bonded-stream counts include 16.

That's very relevant to your architecture.

Your proposed arrangement is:



ONE F-TILE

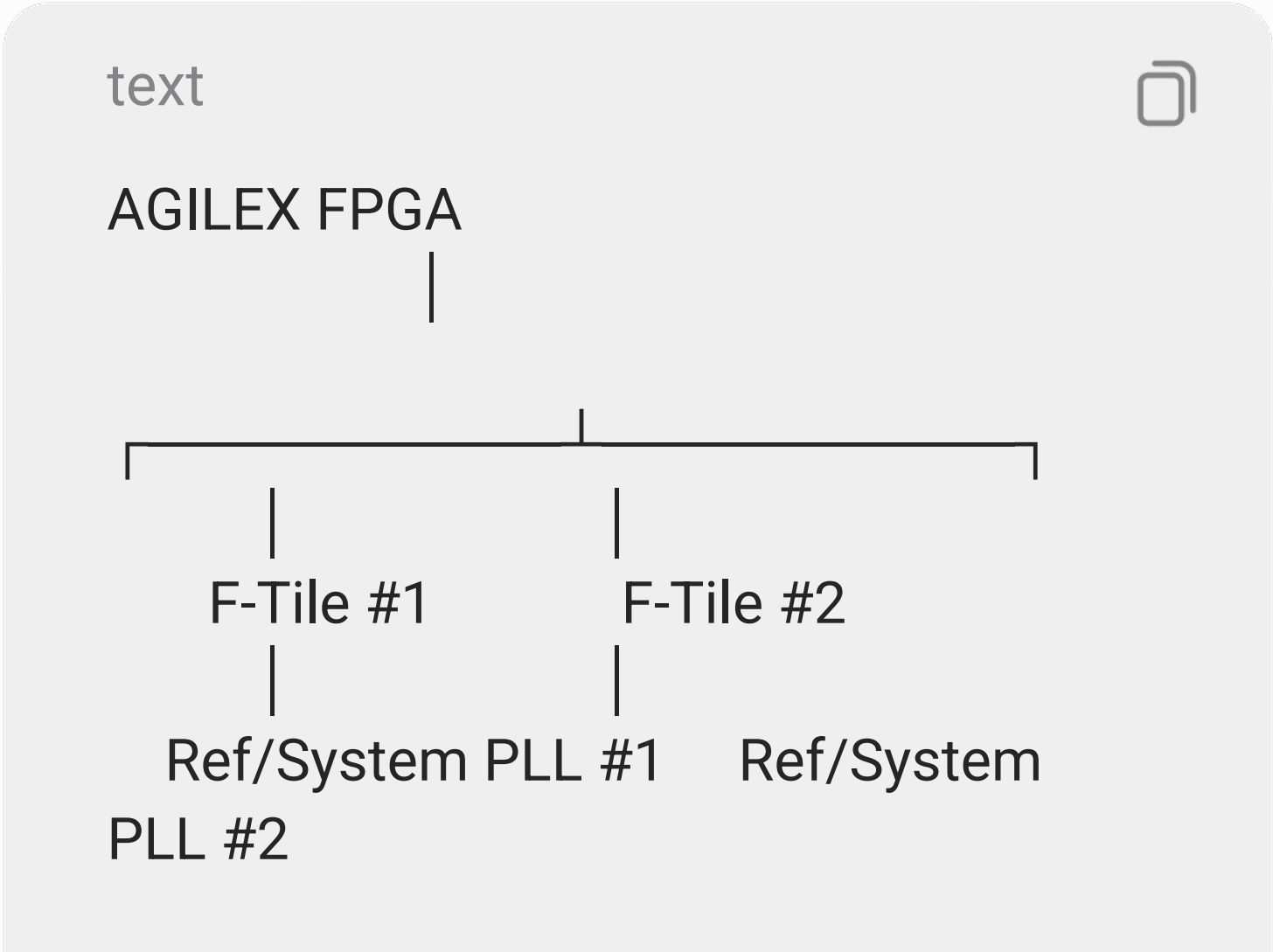


This is the architecture I would investigate in Quartus.

6. Now your TWO F-Tile case

If your Agilex device has two F-Tiles, then each tile has its own F-Tile clocking resources.

For example:



400G #1
8 FGT

400G #2
8 FGT

This is actually a very clean architecture if your board has two 400G interfaces physically associated with two different tiles.

Each F-Tile has its own clocking infrastructure.
So you don't have:

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ONE PLL → 2 different F-Tiles

as the fundamental tile architecture.

Instead, think:

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F-Tile #1

└─ its Reference/System PLL

└─ 400G #1

F-Tile #2

- └─ its Reference/System PLL
- └─ 400G #2

That can be easier from a clocking and placement perspective.

7. What I would recommend for your requirement

Your requirement is:

Only 2 × 400G Ethernet

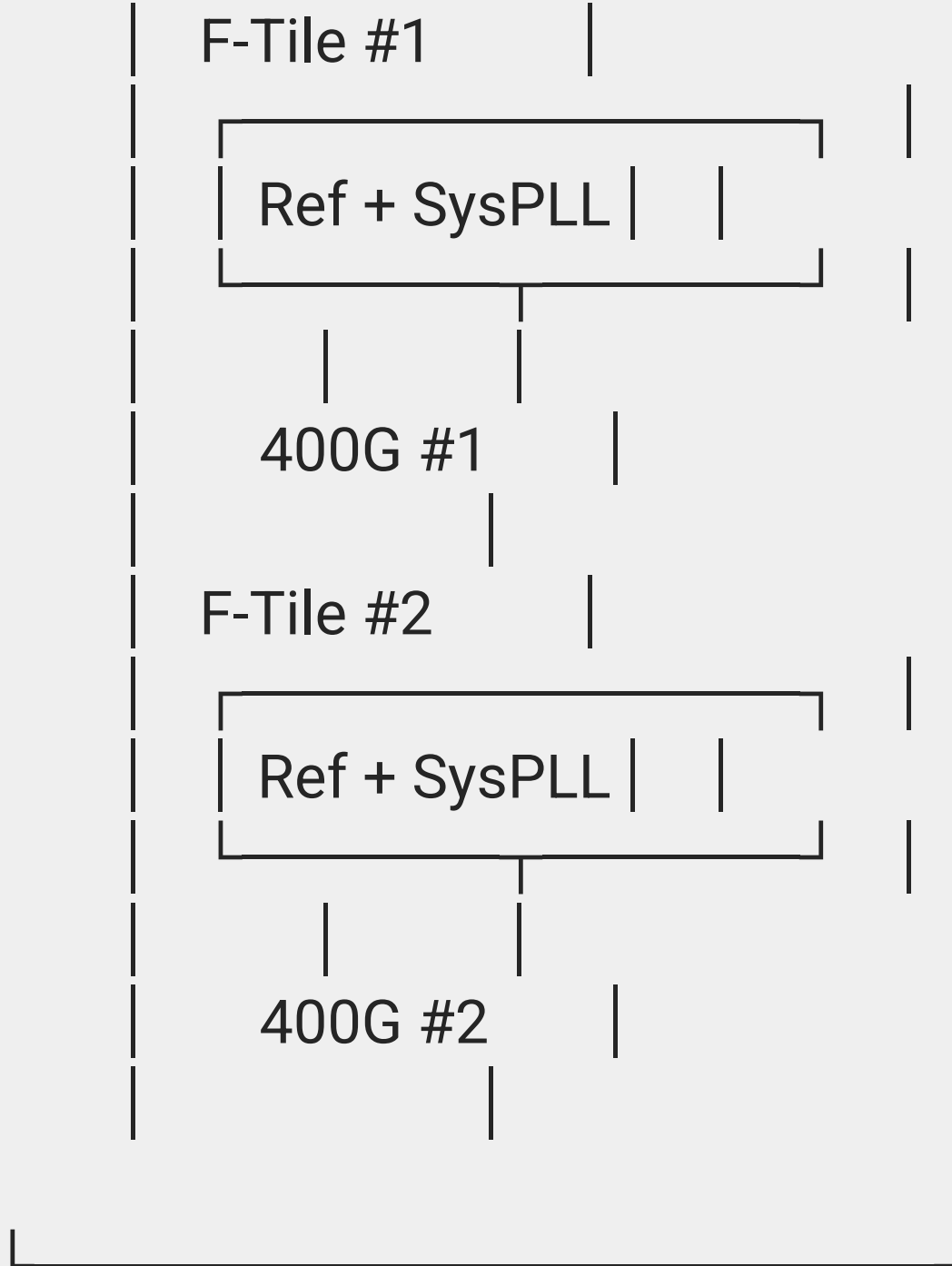
If the FPGA has two F-Tiles available, I would initially investigate this architecture:

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AGILEX 7





This gives you a straightforward one 400G interface per F-Tile arrangement.

However, if your exact OPN has a physical arrangement that makes both 400G ports on one F-Tile more convenient, the documented FGT resource and 400GE-8 mode make that

architecture worth evaluating too.

One correction to our earlier discussion

Don't think of F-Tile's 16 FGT + 4 FHT simply as "20 identical transceivers." Intel explicitly distinguishes them:

16 FGT: up to 58.125 Gb/s PAM4

4 FHT: up to 116 Gb/s PAM4

and not all PMAs necessarily have identical supported rates/bonding/physical availability. So for your 400GE-8 architecture, the 16 FGT side is the key resource.

Bottom line:

2 × 400G is technically achievable with two F-Tiles, and the clocking architecture supports the required multi-port operation. Intel's documentation gives strong direct evidence for the 400GE-8 mode, FGT resources, shared F-Tile clock architecture, and 830.078/415.039 MHz

clocking. The remaining step for a *guaranteed* implementation is checking your exact Agilex 7 OPN + F-Tile placement/channel grouping in Quartus rather than relying only on lane arithmetic.